

MODULE 1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) refers to the science and engineering of making intelligent machines, especially intelligent computer programs that can mimic human cognitive functions such as thinking, learning, and problem-solving.

- **Layman's View:** AI enables machines to mimic human behavior.
- **Researcher's View:** AI is a set of algorithms or instructions that produce results without explicit programming, allowing machines to think and act rationally.
- **Examples:** AI is used in chess-playing computers, autonomous driving cars, and virtual assistants.

Fundamentals of AI

Definition: AI is the study of agents that receive percepts from the environment and perform actions.

The Four Approaches to AI:

1. **Thinking Humanly:** The Cognitive Modeling approach (e.g., mimicking the human brain's decision process).
2. **Acting Humanly:** The Turing Test approach (e.g., a machine that can fool a human into thinking it is human).
3. **Thinking Rationally:** The "Laws of Thought" approach (e.g., using formal logic to solve any problem).
4. **Acting Rationally:** The Rational Agent approach (e.g., acting to achieve the best outcome/goal).

2. How Does AI Work?

AI systems rely on three core cognitive processes:

1. **Learning Processes:** Acquiring data and creating rules (algorithms) to turn that data into actionable information.
2. **Reasoning Processes:** Choosing the right algorithm to achieve a desired outcome.

3. **Self-Correction Processes:** Continuously refining algorithms to improve accuracy over time.

3. Advantages and Disadvantages of AI

Advantages

- **Efficiency:** Performs detailed data tasks quickly and accurately.
- **Speed:** Processes large amounts of data faster than humans.
- **Consistency:** Delivers reliable results without fatigue.
- **24/7 Availability:** Operates continuously without breaks.
- **Optimization:** Better resource utilization and automation of complex processes.

Disadvantages

- **High Cost:** Development and implementation are expensive.
- **Expertise Required:** Needs skilled professionals to build and manage systems.
- **Data Dependency:** Inaccurate or incomplete data can lead to errors.
- **Job Displacement:** May replace certain human roles in the workforce.
- **Ethical Concerns:** Raises issues regarding privacy, bias, and accountability.

4. Types of Artificial Intelligence

Based on Capabilities

1. **Narrow AI (Weak AI):** Designed to perform specific tasks (e.g., Siri, Alexa, recommendation engines).
2. **General AI (Strong AI):** A hypothetical AI with human-like reasoning and adaptability across any domain.
3. **Super AI:** A future concept where AI surpasses human intelligence in all aspects.

Based on Functionality

1. **Reactive Machines:** Responds to current inputs with no memory of past experiences (e.g., IBM Deep Blue).

2. **Limited Memory:** Can learn from historical data to make decisions (e.g., self-driving cars, fraud detection).
3. **Theory of Mind:** A future goal for AI to understand human emotions and social cues.
4. **Self-Awareness:** The most advanced hypothetical form where AI possesses consciousness and self-reflection.

5. Key Subfields: AI, ML, and DL

- **Artificial Intelligence (AI):** The broadest term, encompassing any technique that allows machines to mimic human intelligence.
- **Machine Learning (ML):** A subfield of AI focused on giving machines the ability to learn from data and identify patterns without being explicitly programmed.
- **Deep Learning (DL):** An advanced technique within ML that uses multi-layered neural networks to learn from massive amounts of data (e.g., face recognition).

Defining Intelligence (Gardner's Theory)

Intelligence is the ability to learn, understand, and apply knowledge to solve problems and make decisions.

- **Linguistic:** Effectively speaking and using language.
- **Mathematical:** Thinking logically and solving scientific problems.
- **Spatial:** Visualizing and manipulating objects in space.
- **Musical:** Creating and understanding music and rhythms.
- **Interpersonal:** Understanding other people's feelings and intentions.

1. Components of Intelligence

Intelligence, whether human or artificial, is built from several key components used to stimulate intelligent behavior:

- **Reasoning :** The process of making decisions or predictions based on facts or evidence.
 - **Inductive Reasoning:** Drawing general conclusions from specific examples (e.g., Seeing many white swans assuming all swans are white).

- **Deductive Reasoning:** Drawing specific conclusions from general rules (e.g., All birds have feathers; an eagle is a bird an eagle has feathers).
- Learning : The process of acquiring new knowledge or improving skills through experience.
 - **Auditory Learning:** Learning by hearing (e.g., lectures).
 - **Episodic Learning:** Learning through a sequence of events.
 - **Motor Learning:** Learning through movement (e.g., playing sports).
 - **Observational Learning:** Learning by watching others.
- Problem Solving : Identifying problems and finding suitable solutions.
 - **Steps:** Identifying the issue Exploring possible solutions Selecting and applying the best solution.
- Perception : The ability to sense and interpret information from the environment. AI uses sensors (cameras, microphones, etc.) to gather data.
- Linguistic Intelligence : The capability to use and comprehend verbal and written language effectively.

2. Difference Between Human and AI (Important , Asked in Model QP)

Aspect	Human Intelligence	Machine Intelligence
Origin	Naturally developed through biological evolution.	Artificially created by humans through programming.
Learning Ability	Learning from experience, emotions, and understanding.	Learning from data using algorithms (Machine Learning).
Perception	Perceives using human senses (sight, hearing, etc.).	Perceives data using sensors (cameras, mics) and rules.
Memory & Recall	Recalls information based on associative memory.	Retrieves stored data using search algorithms or databases.
Decision Making	Considers logic, intuition, and emotions.	Decisions based purely on logic, data, and programmed rules.

3. Agent and Environment

- **Agent** : Anything that can perceive its environment through **sensors** and act upon that environment using **effectors/actuators**.
- **Environment** : Everything outside the agent that it interacts with or reacts to.

Types of Agents

1. **Simple Reflex Agent**: Acts only on the current percept, ignoring history.
2. **Model-Based Reflex Agent**: Maintains an internal "model" or state of the world to track objects.
3. **Goal-Based Agent**: Acts to achieve a specific, defined goal (e.g., GPS navigation).
4. **Utility-Based Agent**: Chooses actions based on a "utility" or preference (best satisfaction).
5. **Learning Agent**: Learns from past experiences to improve performance over time (e.g., ChatGPT).

Agent and Environment

- **PEAS Descriptor**: Every agent is defined by its **Performance measure, Environment, Actuators, and Sensors**.

Example (Medical Diagnosis System):

- **P**: Healthy patient, reduced costs.
- **E**: Patient, hospital, staff.
- **A**: Screen display (report/diagnosis).
- **S**: Keyboard (symptom entry), medical sensors.

4. Search Algorithms

Search is the method an AI system uses to navigate through a "problem space" to find a solution.

A. Uninformed Search (Blind Search)

These have no additional information about the goal state beyond the problem definition.

- **Depth-First Search (DFS):** Explores as far as possible along each branch before backtracking. Uses a **Stack (LIFO)**.
 1. **Logic:** Goes deep into one branch before backtracking.
 2. **Data Structure:** Stack (LIFO).
 3. **Pro:** Low memory requirements.
- **Breadth-First Search (BFS):** Explores all nodes at the current depth level before moving to the next. Uses a **Queue (FIFO)**.
 1. **Logic:** Level-by-level exploration.
 2. **Data Structure:** Queue (FIFO).
 3. **Pro:** Finds the shortest path.
- **Uniform Cost Search (UCS):** Expands the node with the lowest path cost.

B. Informed Search (Heuristic Search)

Uses extra information (Heuristics) to find the goal more efficiently.

- **Greedy Best-First Search:** Always expands the node that "appears" to be closest to the goal based on the heuristic value.

Best-First Search (Greedy) Uses a Heuristic Function $h(n)$ (an estimate of the distance to the goal).

It only cares about the immediate "best" looking step.

Formula: $f(n) = h(n)$.

5. Knowledge Representation (KR)

KR is a fundamental area of AI dealing with how to store and manipulate knowledge to enable intelligent decision-making.

Intelligence requires a Knowledge Base (KB). A Knowledge-Based Agent operates through:

1. **TELL:** The agent is informed of a fact.

2. **ASK:** The agent is queried on what to do next.

Types of Knowledge

1. **Declarative Knowledge:** Facts and statements (e.g., "The sky is blue").
2. **Procedural Knowledge:** Step-by-step processes or algorithms (e.g., how to ride a bike).
3. **Meta-Knowledge:** Knowledge about knowledge.
4. **Heuristic Knowledge:** "Rules of thumb" used for problem-solving.
5. **Structural Knowledge:** Relationships between concepts (e.g., hierarchies).

AI vs. Cognitive Computing vs. Augmented Intelligence

1. **AI: Autonomy** (Machine does the work).
2. **Augmented Intelligence: Assistance** (Machine helps the human do work better).
3. **Cognitive Computing: Simulation** (Machine thinks like a human to solve complex, ambiguous problems).